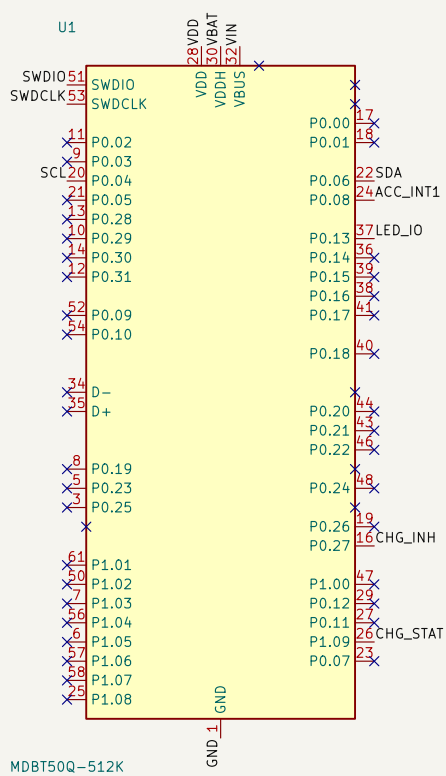
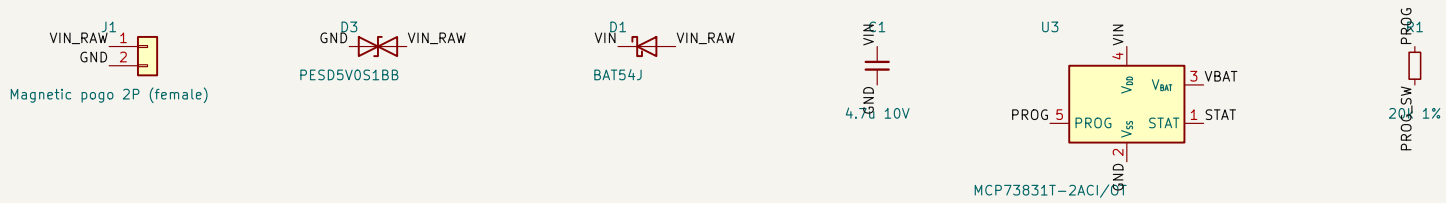


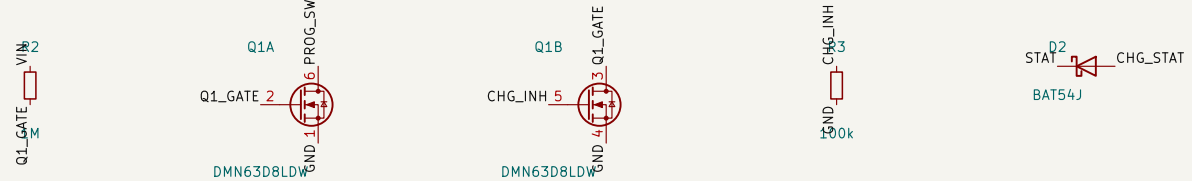
MCU / radio (nRF52833 high-voltage mode)



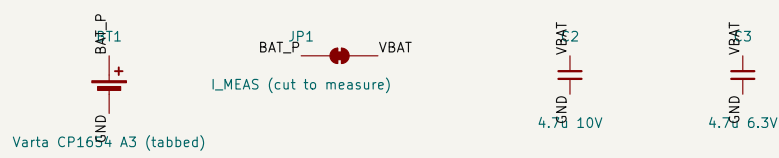
Charging input & charger



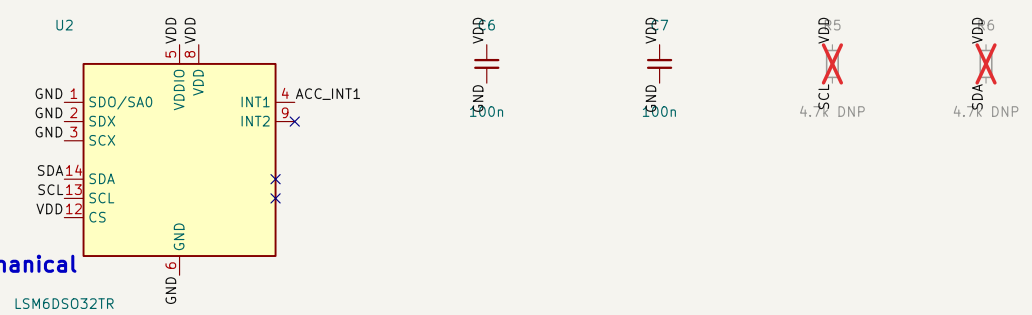
Charge inhibit / status



Battery



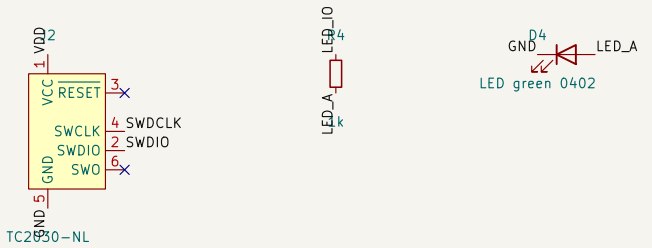
IMU (accelerometer + gyroscope)



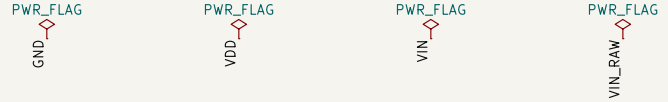
Mechanical



Programming & status LED



Power flags (ERC)



design.py – ShotPuck circuit, single source of truth.

gen_sch.py turns this into shotpuck.kicad_sch and gen_pcb.py into shotpuck.kicad_pcb. Change the circuit here, then re-run both.

Power path

J1 magnetic pogo (VIN_RAW) -> D1 BAT54J (reverse polarity) -> VIN

VIN -> U3 MCP73831-2 (4.20 V, 50 mA via R1 = 20k) -> VBAT (CP1654)

VBAT -> U1 VDDH (nRF52833 high-voltage mode, REG0 -> VDD 3.0 V)

VDD -> U2 LSM6DS032 (VDD + VDDIO), LED, SWD VCC sense

IMU

U2 LSM6DS032 on I2C at 0x6A (SA0 = GND, CS = VDD selects I2C). Aux master pins SDx/SCx to GND; INT2, OCS_Aux, SDO_Aux unconnected.

Charge inhibit (temperature, firmware)

PROG -> R1 -> Q1A drain; Q1A gate pulled to VIN by R2 (charging allowed whenever a cable is present). Q1B (gate = CHG_INH, R3 pull-down) pulls Q1A gate low -> PROG floats -> MCP73831 charge disabled.

Current measurement

JP1 (bridged solder jumper) in series with the cell; cut, insert an ammeter, re-bridge with solder.

Charge status

STAT -> D2 (cathode) ; D2 anode -> CHG_STAT GPIO with pull-up enabled only while VBUS is present (no standby leakage).